

Global Patent Infringement Analysis: US 11,354,503 B2 (Samsung — Gesture-Based Auto-Complete for Handwriting)

TL;DR

- **No literal infringers identified.** After a global survey of stylus/handwriting products from 2017–2026, no commercial product was found that implements the patent's distinguishing limitation — a gesture whose *length* is mapped to the *number of characters* to be predicted in handwriting auto-complete.
- **Three products warrant low-confidence "watch-list" monitoring under the doctrine of equivalents** because they implement adjacent handwriting auto-complete or predictive functions: **Goodnotes 6 "Word Complete"** (discontinued 31 Mar 2025), **MetaMoJi mazec** (handwriting predictive text), and **Apple "Smart Script"** in iPadOS 18 Notes (released 16 Sep 2024). None map gesture length to character count.
- **Samsung itself has not shipped the patented gesture in Samsung Notes, the S Pen suite, or One UI 6/7/8**, so the patent currently functions as a defensive/portfolio asset rather than a product feature — and so far appears uncited in commercial competitor implementations.

Key Findings

1. The patent's novel limitation is a narrow geometric/semantic mapping. Independent Claims 1 and 11 require that "the length of the at least one gesture corresponds to a number of characters to be determined in the at least one remaining text." This is the differentiator from generic handwriting-autocomplete prior art (Bi et al. 2014, Ouyang 2014, Mazec/MyScript predictive UIs) and is the limitation any infringement claim chart must establish. (uspto)

2. The corresponding academic paper confirms the inventors' own characterization of novelty. Sahu, Mitra, Singh, Veera (Samsung R&D Institute Bangalore, per the Semantic Scholar record) & Venkatesan, *Gesture-Based Auto-Completion of Handwritten Text*, MobileHCI '19, DOI 10.1145/3338286.3340136, abstract verbatim: "we propose a gesture-based solution to recognize and complete partially-written words, where the user indicates to the system the missing text length by a stroke gesture, thus ruling out a huge subset of possibilities. [...] using a gesture stroke length based searching method not only reduced the processing time by 27% but also showed an accuracy increase of 10%." The paper self-cites the same patent application (US App. 16/047,537), confirming it is the academic counterpart. (ACM Digital Library + 2)

3. Samsung Notes / S Pen does not implement the patented feature. Reviewed Samsung support pages for One UI 4.1–8 across US/SG/CA/AE/UK markets: documented S Pen handwriting capabilities are limited to convert-to-text, Clean up / Align handwriting, Handwriting help (letter-shape guidance), Note Assist (Auto format / Summarize / Spell-check / Translate — all post-conversion typed-text features under Galaxy AI), shape recognition (auto-fix), stroke/area eraser, S Pen-to-text in input fields, and shape "hold-pen" auto-regularize. None of these involve drawing a stroke of varying *length* to indicate the *number* of remaining characters. The patented technique therefore appears unimplemented by its own assignee.

4. Closest functionally adjacent commercial product: Goodnotes 6 "Word Complete" (Aug 2023 – 31 Mar 2025). Per TechCrunch (Ivan Mehta, 9 Aug 2023): "After users write a minimum of three characters, the app will suggest potential word completions. In an interview with TechCrunch, Chan said the company aims to offer full word prediction in the future." Suggestions appear in a pop-up the user taps to insert. There is **no gesture-length input** — completion is triggered by the prefix and on-device ML model alone, with no horizontal/wavy continuation stroke that encodes a character count. Goodnotes' own blog (goodnotes.com/blog/introducing-goodnotes-6) confirms: "*Update from March 2025: We're discontinuing the Word Complete experimental feature on March 31st. After a year of testing and your valuable feedback, we've learned that other features (like Smart Ink, Ask Goodnotes, and audio transcription) are more impactful for our users.*" **Verdict: NOT a literal infringer; doctrine-of-equivalents argument weak because the "substantially the same way" prong fails** — Word Complete uses a passive list-based UI, not the patented "gesture as input signal" mechanism.

[TechCrunch](#) [Goodnotes](#)

5. MyScript Nebo / MyScript Notes (renamed Sept 2025) — no length-based gesture.

MyScript's published gesture set is type-coded, not length-coded: Erase (scratch-out), Join (upward line), Break (downward line), Underline, Strike-through, Highlight (frame). None map stroke length to character count. **Verdict: NOT infringing.** [MyScript Help](#) [Myscript](#)

6. Apple Scribble (iPadOS 14+) and Smart Script (iPadOS 18, released 16 Sep 2024 per Apple Newsroom: "iPadOS 18 is available today, bringing more ways to customize the Home Screen and Control Center, along with new Smart Script handwriting tools in Notes") — no length-based mapping. Apple's gestures are also type-coded: scratch wavy = delete, vertical line between chars = join/insert space, touch-and-hold = open insert space, circle/underline = select. Smart Script adds auto-refinement of the user's handwriting style and on-device autocomplete suggestions, but none of these encode a character count via gesture length. **Verdict: NOT infringing.** [Apple + 3](#)

7. Gboard stylus handwriting (Pixel Tablet, Galaxy Z Fold, Android 14+), Microsoft Windows Ink / OneNote, Huawei FreeScript / M-Pencil, reMarkable Type Folio, Onyx Boox, Amazon Kindle Scribe — all reviewed for gesture-driven auto-complete with stroke-length-to-character mapping; none implement the feature. Predictive completion that exists (e.g., OneNote tab-to-complete, Gboard word suggestions, Huawei FreeScript) is based on context/lexicon, not gesture length. **Verdict: NOT infringing.**

8. MetaMoJi mazec / mazec3 (MyScript-licensed engine). Per BusinessNewsDaily ("Top 10 Handwriting Recognition Apps"): "As you begin to write, Mazec displays predictive suggestions and phrases to choose from – you usually don't have to write out an entire word before the app completes it. Mazec intelligently detects your choices, learns specific phrases and even offers emojis." This is prefix-based ranking, with the user tapping a candidate from a list. There is no "draw a stroke whose length equals the character count" mechanism. **Verdict: NOT infringing.**

[Business News Daily](#)

9. Forward citations / patent-family relevance. A search for forward citations of US 11,354,503 B2 returned no commercial competitor patents that cite this Samsung disclosure. Meta

Platforms' US20220050696A1 ("Auto-completion for Gesture-input in Assistant Systems") is conceptually adjacent but covers VR/AR camera-detected *hand gestures* for assistant-system intent prediction, not handwriting stroke-length-to-character mapping, so it is not a relevant competitor implementation. [Google Patents](#)

10. Validity caveats from Samsung's own prosecution record. The patent's specification cites prior art including Bi et al. 2014 and Ouyang 2014 on handwriting prediction generally. The narrow gesture-length-encodes-character-count limitation is what survived examination and is precisely the limitation any infringer must meet — there is essentially no doctrine-of-equivalents "wiggle room" because if a product uses a completion gesture but the length is irrelevant, the "substantially the same way" prong fails.

Details

Patent claim breakdown (independent Claim 1 elements)

#	Claim element	Required mapping
1a	Detecting at least one gesture performed by a user to complete an incomplete text	Gesture must be the trigger
1b	Determining a length of the at least one gesture	System must quantify length
1c	Determining at least one remaining text ... based on the length of the at least one gesture and the incomplete text	Length + prefix → prediction
1d	The length of the at least one gesture corresponds to a number of characters to be determined in the at least one remaining text	The novel limitation
1e	Forming at least one complete text by adding the at least one remaining text to an end of the incomplete text	Concatenation
1f	Displaying the at least one complete text	Output

Claim 11 mirrors Claim 1 in apparatus form (memory + display + processor). Dependent claims add: scanning for limiter characters (cl. 2), gesture-type→tense/verb-form (cl. 3, e.g., upward vs. downward stroke for past/future tense), air gesture without touching display (cl. 7), and marking gestures (e.g., box around word) for thesaurus alternatives (cl. 8). [uspto](#)

Product-by-product infringement assessment

Apple Scribble / Smart Script (iPadOS 14 - iPadOS 18, latter released 16 Sep 2024 per Apple Newsroom)

- Element 1a: ✓ Gestures detected (scratch, circle, vertical line)

- Element 1b: X No published evidence Apple computes "length" as an input variable for completion
- Element 1d: X No character-count mapping
- **Severity: NONE. Confidence: HIGH.** Evidence: Apple Support 117716; Apple Support guide355ab2a7; Apple Newsroom 16 Sep 2024; AppleInsider Sep 2020; Creative Bloq.

Goodnotes Ltd. — Goodnotes 6 "Word Complete" (Aug 2023, sunset 31 Mar 2025)

- Element 1a: X Trigger is prefix length (≥ 3 chars per TechCrunch 9 Aug 2023), not a gesture
- Element 1d: X No length-to-character-count mapping
- **Severity: NONE (literal). Doctrine of equivalents: LOW.** Confidence: HIGH. Evidence: Goodnotes blog "Introducing Goodnotes 6" (incl. March 2025 discontinuation note); MacRumors 9 Aug 2023; TechCrunch 9 Aug 2023 (Ivan Mehta); Goodnotes Support "Sunsetting Word Complete."

MyScript Nebo / MyScript Notes / iink SDK

- Element 1a: ✓ Gestures detected
- Element 1d: X Gestures are *type-encoded* (scratch=erase, $\uparrow$ =join, $\downarrow$ =break, frame=highlight); length is not used for character count
- **Severity: NONE. Confidence: HIGH.** Evidence: help.myscript.com/nebo/edit-content/pen-gestures-and-interactions; myscript.com/notes; Paperlike review.

Samsung Electronics — Samsung Notes / S Pen / Note Assist (One UI 6/7/8) (assignee's own product)

- No gesture-length-to-character-count completion documented anywhere
- **Severity: N/A (Samsung is the patentee).** Confidence: HIGH. Evidence: samsung.com/us/support/answer/ANS10003634; ANS10000941; ANS10002017; eu.community.samsung.com 14150714.

Google LLC — Gboard stylus handwriting (Pixel Tablet / Galaxy / Android 14+)

- Element 1a: ✓ ; Element 1b: X no length input; Element 1d: X
- Gestures are type-coded (strike to select/delete; circle to select; flick for line break)
- **Severity: NONE. Confidence: HIGH.** Evidence: 9to5Google 17 Dec 2023; Android Central; Google Support gboard/9108773.

Microsoft Corp. — OneNote / Windows Ink

- Predictive text on typing only; pen gestures are highlight/scribble-to-erase; no gesture-length-based completion
- **Severity: NONE. Confidence: HIGH.** Evidence: How-To Geek "OneNote on Windows Gets Improved Pen and Ink Gestures"; BusinessNewsDaily OneNote review.

- Documented stylus gestures: select (horizontal line through text), delete (wavy line through word), space management (vertical line); these *follow* recognition rather than driving multi-character completion ([Huawei Mexico](#))
- **Severity: NONE. Confidence: HIGH.** Evidence: consumer.huawei.com en-us15879071 (FreeScript), en-us15821589 (usage guide); HUAWEI MatePad Paper user guide.

MetaMoJi — mazec / mazec3 (uses MyScript engine)

- Predictive completion based on prefix and learned dictionary (per BusinessNewsDaily: "you usually don't have to write out an entire word before the app completes it"); user taps suggestion; no length-encoding stroke ([Business News Daily](#))
- **Severity: NONE. Confidence: HIGH.** Evidence: product.metamoji.com/en/mazec; App Store id943711253; BusinessNewsDaily.

reMarkable AS — reMarkable 2 / Paper Pro (Type Folio + handwriting conversion)

- Handwriting → text conversion, no autocomplete-via-gesture
- **Severity: NONE. Confidence: HIGH.** Evidence: remarkable.com/using-remarkable/improve-your-workflow/combine-handwriting-and-typing-with-type-folio.

Onyx Boox / Supernote / Kindle Scribe / Kobo Elipsa

- All implement handwriting-to-text conversion but do not have a documented gesture-length-to-character-count completion feature.
- **Severity: NONE. Confidence: MEDIUM** (less documentation available — based on absence of evidence across product feature pages and reviews).

Forward citations / competitive patent landscape

A search for forward citations of US 11,354,503 B2 did not surface any commercial competitor patents directly citing this disclosure. Meta Platforms' US20220050696A1 ("Auto-completion for Gesture-input in Assistant Systems") is conceptually adjacent but covers VR/AR camera-tracked hand gestures for assistant intent, not handwriting; it is not a relevant infringement candidate. The closest publication universe is academic (the inventors' own MobileHCI '19 paper plus the prior art cited therein — Bi et al. 2014, Ouyang 2014, Xing et al. 2015), none of which is a commercialized infringing product. ([uspto](#))

Company / Product	Claim(s) potentially read on	Severity	Confidence	Evidence count	Key evidence
Apple Inc. — Scribble / Smart Script (iPadOS 14 - 18, 16 Sep 2024)	Claims 1, 11 — fails element 1d	NONE	HIGH	5	Apple Support 117716; Support guide355ab2a7; Apple Newsroom 16 Sep 2024; AppleInsider 2020; Creative Bloq
Goodnotes Ltd. — Goodnotes 6 "Word Complete"	Claims 1, 11 — fails 1a/1d (no gesture); weak DoE	LOW (DoE only)	HIGH	4	Goodnotes blog "Goodnotes 6" (incl. Mar 2025 sunset); MacRumors 9 Aug 2023; TechCrunch 9 Aug 2023 (Ivan Mehta); Goodnotes "Sunsetting Word Complete"
MyScript SAS — Nebo / MyScript Notes / iink SDK	Claims 1, 11 — gestures are type-coded, not length-coded	NONE	HIGH	3	help.myscript.com pen-gestures; myscript.com/notes; Paperlike review
Samsung Electronics — Samsung Notes / S Pen	N/A (patentee)	N/A	HIGH	4	samsung.com support ANS10003634, ANS10000941, ANS10002017; eu.community.samsung.com 14150714
Google LLC — Gboard stylus handwriting	Claims 1, 11 — type-coded gestures, no length mapping	NONE	HIGH	3	9to5Google 17 Dec 2023; Android Central; Google Support 9108773
Microsoft Corp. — OneNote / Windows Ink	Claims 1, 11 — predictive text not gesture-driven	NONE	HIGH	2	How-To Geek OneNote ink; BusinessNewsDaily OneNote review

Company / Product	Claim(s) potentially read on	Severity	Confidence	Evidence count	Key evidence
Huawei — FreeScript / M-Pencil / HUAWEI Notes	Claims 1, 11 — gestures are editing, not completion	NONE	HIGH	3	consumer.huawei.com en-us15879071; en-us15821589; MatePad Paper guide
MetaMoJi — mazec / mazec3 (MyScript engine)	Claims 1, 11 — prefix-based predictive list, no gesture	NONE	HIGH	3	product.metamoji.com/en/mazec; App Store id943711253; BusinessNewsDaily
reMarkable AS — reMarkable 2 / Paper Pro	Claims 1, 11 — no autocomplete feature	NONE	HIGH	2	remarkable.com Type Folio; remarkable.com Paper Pro
Onyx Boox / Supernote / Kindle Scribe / Kobo Elipsa	Claims 1, 11 — no documented matching feature	NONE	MEDIUM	1	Product feature pages, absence of evidence
Lenovo / Xiaomi / OPPO- OnePlus stylus apps	Claims 1, 11 — no documented matching feature	NONE	MEDIUM	0-1	No matching feature surfaced in review

Recommendations

Stage 1 — Immediate (next 30 days):

1. Do **not** open commercial-licensing or enforcement campaigns; the patent reads on a very narrow gesture-length-to-character-count mapping that no current product implements. The risk of a 35 USC §285 attorney-fee award for chasing weak claims outweighs upside.
2. Manually verify Google Patents' "Cited By" list directly at patents.google.com/patent/US11354503B2/en to confirm forward-citation count (automated search returned none, but Google Patents' live "Cited By" page should be checked by counsel).
3. File a portfolio note that the patent is best deployed defensively or as a cross-licensing chip in negotiations with Apple, Microsoft, or Google.

Stage 2 — Watch-list monitoring (90-day cadence): 4. Re-monitor **Goodnotes** for any successor to "Word Complete" — the discontinuation on 31 Mar 2025 was attributed to feature impact, but a future re-launch that introduces a gesture-input trigger should be re-evaluated. 5. Re-monitor **Apple Smart Script / Apple Intelligence on iPadOS 19+** — if Apple adds a Pencil-stroke continuation gesture (e.g., a "swipe to finish word" tied to Apple Intelligence writing tools), it could move from NONE to MEDIUM. 6. Re-monitor **Microsoft Copilot in OneNote** — Copilot integration could plausibly introduce gesture-driven completion in future Surface Pen releases. 7. Re-monitor **MyScript iink SDK changelogs** — MyScript licenses to numerous OEMs and is the most likely vendor to add a length-based completion gesture as a differentiator.

Triggers that would change the recommendation:

- Discovery of any product where drawing a *longer* horizontal/wavy continuation stroke yields *more characters* in a single suggested word → escalate to claim chart and demand letter.
- Forward citation of US 11,354,503 B2 by any Apple, Google, Microsoft, MyScript, or Huawei patent application → escalate to FTO review for that assignee.
- Samsung's own commercialization of the feature in a future One UI release → marks the patent as a productized asset and increases settlement leverage.

Caveats

- **Negative-evidence inference.** Several of the "NONE — MEDIUM confidence" determinations (Onyx, Supernote, Kindle Scribe, Kobo, Lenovo, Xiaomi, OPPO/OnePlus) rest on absence of any documented matching feature in feature pages and reviews; an undocumented "hidden" feature is possible but unlikely given marketing incentives to publicize novel ink features.
- **Patent validity caveat.** Samsung's own specification cites Bi et al. 2014 and Ouyang 2014 as prior art on handwriting prediction. The valid scope is limited to the gesture-length-encodes-character-count limitation; any attempt to read Claim 1 on general handwriting-autocomplete UI (e.g., Goodnotes' Word Complete pop-up) would likely fail under 35 USC §112 written-description and §103 obviousness scrutiny.
- **Forward-citation search was tool-limited.** The subagent could not retrieve Google Patents' "Cited By" list programmatically; the no-citing-patent finding is provisional and counsel should verify on the live patents.google.com page.
- **Geographic scope.** This analysis covers globally distributed English-language documentation and US/EU/Asia commercial products. Chinese-domestic-only handwriting apps (Kilonotes, Penly, Honor stylus apps) may have features not captured in English-language sources; a Chinese-language follow-up review is advisable before any China-market assertion.
- **Claim construction caveat.** "Length" in Claim 1 could plausibly be construed under *Phillips* to mean either (i) physical pixel/centimeter length or (ii) duration; the patent specification

primarily teaches physical length but a defendant could argue (ii) creates ambiguity. Any litigation would need an early Markman position.

- **Doctrine of equivalents narrowing.** Because the prosecution history likely added the "length corresponds to a number of characters" limitation to overcome prior-art rejections (a common pattern in HWR patents), prosecution history estoppel may sharply limit any DoE expansion. Pre-suit, review the file wrapper for amendments.
- **Speculation flagged.** Statements about future product directions (Apple Intelligence, Copilot-in-OneNote, MyScript SDK roadmap) are forward-looking and not established facts; they are watch-list signals only.
- **Inventor affiliation precision.** Per the Semantic Scholar record for DOI 10.1145/3338286.3340136, Viswanath Veera is listed as "Samsung R&D Institute Bangalore"; the other co-authors are within the Samsung R&D India group, consistent with the patent's Indian-priority application 201741026780.